OBPDS 2.0: a verified one-dimensional Schrödinger–Poisson and drift–diffusion solver for III–V heterostructures

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Abstract

We describe OBPDS 2.0, an open-source program that solves the one-dimensional electrostatic, transport and envelope-Schrödinger problems of III–V heterostructures on a single grid. It computes self-consistent band diagrams, terminal currents, capacitances and quantised subband structure for layer stacks that may be pseudomorphically strained. It is a modern rewrite of the Open Band Parameters Device Simulator, which stopped at the zero-current approximation and produced no current; the rewrite adds drift–diffusion transport, self-consistent quantisation and strain, and keeps the non-parabolic multi-valley statistics of the original.

The numerical content is what we report. Scharfetter–Gummel fluxes are written in a generalised-potential form that is an identity for arbitrary statistics and, after a cancellation-free rearrangement, returns an identically zero equilibrium current instead of a difference of large numbers. In III–V materials the carrier densities inside one device span more than twenty decades, so the discrete current in a quasi-neutral region is below the round-off of the stored quasi-Fermi levels; the program therefore reads the terminal current from the depletion region and reports the relative deviation of the worst link, which tells the user when the value is arithmetic noise. Quantisation uses the BenDaniel–Duke operator on the same grid, and strain a closed-form Bir–Pikus Hamiltonian with spin–orbit coupling.

Every model is held to closed-form results by 196 assertions, including negative controls that require the correction to vanish where nothing confines.

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A representative bias point converges in two Gummel sweeps and about 1.5 s.

PROGRAM SUMMARY

Program Title: OBPDS 2.0 — Open Band Parameters Device Simulator
CPC Library link to program files: (to be added by Technical Editor)
Developer's repository link: https://github.com/<owner>/obpds
Licensing provisions: AGPL-3.0-or-later. This program is a derivative of OBPDS, which is distributed under the GNU Affero General Public License v3.0; the licence is therefore retained and cannot be replaced by one of the licences on the standard list.
Programming language: Python 3 (>= 3.10); JavaScript/HTML for the browser front end
Supplementary material: Test suite (196 assertions against closed-form solutions), a comprehensive example run with its expected output, the scripts that regenerate every figure and table of this paper, and the web front end.
Nature of problem (approx. 50–250 words): One-dimensional band-diagram and terminal-current calculations for III–V heterostructures couple a strongly nonlinear Poisson equation to degenerate, non-parabolic carrier statistics and to confinement in wells a few nanometres wide. The dynamic range is extreme: in GaAs at room temperature the intrinsic density is 1.7×10^6 cm^{-3} against a doping of 10^{17}–10^{18} cm^{-3}, so electron and hole densities differ by more than twenty decades inside one device. A solver must stay accurate where the current carried by a quasi-neutral region corresponds to a quasi-Fermi gradient five orders of magnitude below the round-off of the stored levels, and must distinguish genuine confinement from a discretisation artefact. Open implementations either stop at the zero-current approximation, in which the quasi-Fermi levels are prescribed and no current can be computed, or cannot be inspected and modified.
Solution method (approx. 50–250 words): Finite-volume discretisation of $\nabla\cdot(\varepsilon\nabla\psi)$ on a uniform grid with the face permittivity taken as the harmonic mean of adjacent nodes, so that the $\varepsilon'\psi'$ term is retained at heterointerfaces. The Poisson system is solved by Newton's method, damped along Bank–Rose lines [3], with every row equilibrated before the solve. Drift–diffusion uses Scharfetter–Gummel fluxes [1] in a generalised-potential form that is exact for any statistics and, rearranged to be cancellation-free, gives an identically zero equilibrium current. Continuity is solved by Newton with the same damping and coupled to Poisson by Gummel iteration [2] from an analytic minority-carrier profile, so that a bias point converges in two or three sweeps from a cold start. Confinement uses a BenDaniel–Duke tridiagonal eigenproblem [6] with harmonic averaging of $1/m^*$, populated as two-dimensional subbands plus a continuum remainder. Pseudomorphic strain enters through a closed-form Bir–Pikus Hamiltonian [5] with spin–orbit coupling; band parameters

follow [4].

Additional comments including restrictions and unusual features (approx. 50–250 words): Restrictions: one-dimensional, single temperature, drift–diffusion with constant low-field mobilities, no hole quantisation, no exchange–correlation, no Auger recombination, and no strain relaxation (the Matthews–Blakeslee thickness is reported, not applied). Quantisation is applied at equilibrium only: there is no quantum-corrected transport, so every current and capacitance is classical. Unusual features: (i) the terminal current is read from the depletion region and reported together with `J_spread`, the relative deviation of the worst discrete link, so that a user can see when the reported value is noise; (ii) the equilibrium current is exactly zero rather than merely small; (iii) every physical model is exercised against closed-form results, including negative controls; (iv) a browser front end serves the same solver, so no installation is needed for classroom use; (v) the program flags the cases in which its own model is invalid, for example a layer thicker than its critical thickness; (vi) band parameters can be calibrated per layer, which lets a reader test what a different valence-band offset would do without editing the parameter tables.

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1. Introduction

One-dimensional band diagrams are the working language of III–V device design. A layer stack, a doping profile and a substrate are enough to answer the questions that come up first: where the electrons sit, how the bands line up, how much of the gap is taken by the conduction-band offset, how a quantum well shifts when it is strained. Tools of this kind have existed for

decades. Bandprof [1] and its descendants, including the Open Band Parameters Device Simulator (OBPDS) [2] that this paper starts from, solve the nonlinear Poisson equation for a prescribed pair of quasi-Fermi levels—the *zero-current approximation*—and draw the result [3]. That approximation is legitimate for a band diagram and useless for a current–voltage curve: with the quasi-Fermi levels fixed there is no equation from which a current could be obtained.

The gap matters because the interesting III–V structures are exactly the ones in which the two effects the zero-current approximation separates are entangled. A modulation-doped heterojunction puts its electrons in a notch a few nanometres wide, where confinement shifts the charge centroid behind the interface and changes the sheet density that a classical calculation predicts [4, 5]. A pseudomorphically strained well has its valence band split by tens of meV, which moves the band edges and hence the confinement [6, 7]. In both cases the confined charge shapes the potential that confines it, so the electrostatics, the statistics and the eigenvalue problem have to be solved together, and a transport calculation has to be added on top if a current is wanted.

Commercial device simulators do all of this, and do it in two and three dimensions. What is scarce is a one-dimensional implementation that is small enough to read, open enough to modify, and verified against results that do not come from itself. Verification is the part that is usually missing: a solver that agrees with a previous version of itself, or with a figure in a textbook, has demonstrated very little. The claim we want to be able to make is narrower and stronger—that each physical model in the program reproduces a closed-form result in the limit where one exists, and that the limits of the model are stated.

This paper describes OBPDS 2.0, a rewrite of OBPDS in Python 3 with three capabilities the original did not have: drift–diffusion transport, a self-consistent Schrödinger–Poisson correction, and pseudomorphic strain. The rewrite also corrects several defects of the original that are worth naming because they are easy to reproduce. The original discretised $\varepsilon\psi'' = -\rho$ and thereby dropped the $\varepsilon'\psi'$ term, which produces a spurious field at every heterointerface with a permittivity contrast. It assigned the applied bias to the wrong quasi-Fermi level, so that its forward-bias examples were reverse biased. And its Shockley–Read–Hall term used the local equilibrium densities where the trap-level densities belong, which inflates the generation rate in the depletion region and suppresses the reverse current by more than an order

of magnitude while leaving the forward characteristic apparently correct.

The nearest contemporary open implementation is Aestimo 1D [8, 9], a Python package that solves the Schrödinger–Poisson problem for layered heterostructures and, in its current release, also offers drift–diffusion routes, a desktop graphical interface and a wider material database that includes the nitrides. The overlap is real—both codes are one-dimensional, both couple the same three equations, and a reader is entitled to ask which to use. Three differences decide it. Aestimo's model documentation states that the current combined electron–hole Schrödinger–Poisson routine rejects the non-parabolic and exchange selections that its older labels refer to, so the standard route is parabolic, with a separate multiband $k \cdot p$ route for those who need more; here Kane non-parabolicity and the $\Gamma/L/X$ valleys stay active throughout. Its documentation describes no strain model, whereas here the substrate is a device-level attribute that moves every band edge. And the treatment of the current differs in kind rather than degree: the cancellation-free flux of Section 4.2 and the reported spread of the discrete current are what this paper is about. Neither code contains the other—Aestimo's material coverage and its desktop interface are broader than ours, and a user who needs nitrides or a multiband $k \cdot p$ route should take it.

The contributions of the present work are the following.

1. A drift–diffusion formulation for non-Boltzmann statistics. The Scharfetter–Gummel flux is written in terms of a generalised potential that is an identity for any statistics, and rearranged so that the two large terms that cancel at equilibrium are eliminated analytically rather than numerically. The equilibrium current is then exactly zero, and the same expression carries the band offsets, the density-of-states gradients and the degeneracy correction without separate terms.
2. An explicit treatment of the numerical difficulty that III–V materials create. Because the minority density can be twenty decades below the majority density, the discrete current in a quasi-neutral region is below the round-off of the stored quasi-Fermi levels. The program reads the terminal current where both carriers are minorities, and reports the relative spread of the discrete current as a diagnostic instead of presenting a noise-dominated number as a result.
3. A quantisation scheme whose discretisation details are stated and tested: the ghost-node flux that distinguishes a Dirichlet wall from a Neumann one, the nodal-sum normalisation that the discrete operator

is self-adjoint under, and harmonic averaging of $1/m^*$ at an interface. Each of these, done the obvious way, produces a spectrum that looks plausible and is wrong.

4. A closed-form Bir–Pikus treatment of the valence band with spin–orbit coupling, together with the three identities that make it checkable, and a strain model that reports the Matthews–Blakeslee critical thickness of every layer and flags the ones that exceed it.
5. A verification suite of 196 assertions against closed-form solutions, including negative controls: a classical p–n junction must come back unchanged when the quantum correction is switched on, a layer on its own substrate must not be strained, and the equilibrium current must vanish identically.

Section 2 describes the program and its interface. Section 3 states the physical models, Section 4 the discretisation and the solvers, and Section 5 the verification strategy and its results. Section 6 shows what the program produces for three structures, Section 7 reports timings and the limits of the numbers, and Section 8 concludes.

2. Program structure and interface

2.1. Layout

The solver is 3064 lines of Python in thirteen files, the browser front end 1744 lines of JavaScript, HTML and CSS, and the verification scripts 929 lines. Table 1 lists the modules and what each one owns. The division is by physics: material parameters, strain, statistics, discretisation, linear algebra and transport are separate, and no module reaches across more than one of those boundaries. Figure 1 shows the data flow.

2.2. Interface

A calculation is described by a list of layers and two contacts. Doping is signed—positive for acceptors, negative for donors—so that a p–n junction reads as written on the wafer:

```
from obpds import Layer, OhmicContact, TwoTerminalDevice

dev = TwoTerminalDevice(
    [Layer(1000, "GaAs", Na=1e17),      # thickness in nm
```

Table 1: Modules of the solver. Line counts include docstrings and blank lines; they indicate scale, not merit.

Module	Responsibility	Lines
`solver.py`	finite-volume Poisson, statistics coupling, C–V	478
`transport.py`	Scharfetter–Gummel fluxes, continuity, Gummel	625
`device.py`	public API: layers, contacts, sweeps, summaries	334
`strain.py`	biaxial strain, Bir–Pikus edges, critical thickness	278
`quantum.py`	BenDaniel–Duke eigenproblem, subbands, continuum	270
`newton.py`	damped Newton, scalar Newton, row equilibration	223
`materials/`	binary and ternary parameter tables, alloys	504
`layer.py`	layer, graded layer, compound layer, contacts	161
`fermi.py`	generalised Fermi–Dirac integrals	146
`web/`	FastAPI service and browser front end	1744
`tests/`	verification suites	929

```
     Layer(1000, "GaAs", Nd=1e17)],
    [OhmicContact(), OhmicContact()],
    T=300, N=1000, approx="kane")

sol = dev.get_equilibrium()             # band diagram
dd  = dev.get_drift_diffusion(0.8)      # dd.J in A/cm^2
V, J = dev.get_iv(-0.5, 1.0, 31)        # I--V sweep
V, C = dev.get_cv(-1.0, 0.3, 41)        # C--V sweep
```

Pseudomorphic strain is a property of the wafer, not of a layer, so it is given once at the device level and applied to every layer:

```
dev = TwoTerminalDevice([Layer(20, "GaAs", Nd=1e17),
                         Layer(8, "GaInAs", composition=0.8),
                         Layer(20, "GaAs", Nd=1e17)],
                        substrate="GaAs")
sol = dev.get_quantum_equilibrium(n_states=8)
sol.subbands["levels_eV"]           # bound levels, eV
sol.subbands["sheet_density_cm2"]   # 2DEG sheet density
```

Band parameters can be calibrated per layer when the tabulated values do not describe a particular sample:

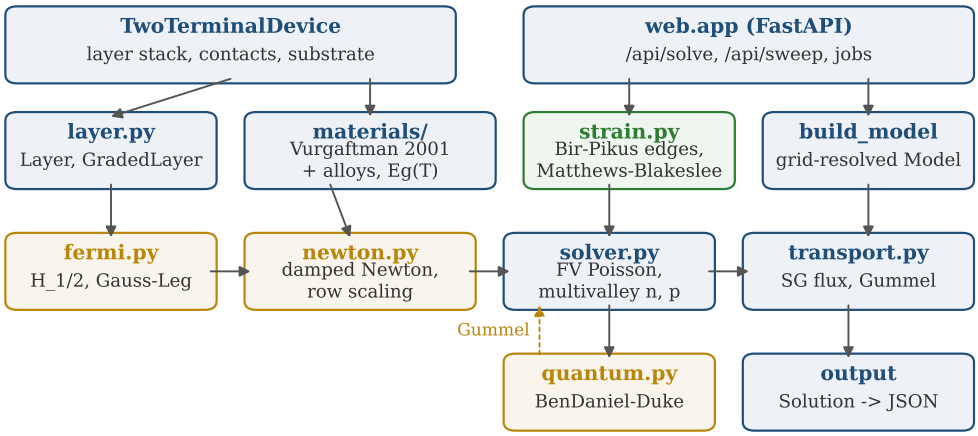


Figure 1: Module map and data flow. A device description is flattened onto the grid once (`build_model`), optionally after strain has moved the band edges; the three solvers then share that array set. Dashed arrow: the quantised density is held fixed while Poisson is re-solved, which is the Gummel loop of Section 3.4.

```
well = Layer(8, "GaInAs", composition=0.8,
             overrides={"VBO": -0.42, "b_shear": -1.9})
```

The valence-band offset and the strain parameters are the two things open to override; the value replaces the unstrained parameter, so strain is applied on top of it. Everything else feeds derived quantities—n_i, N_c, N_v, the non-parabolicity—that would be left stale if half of a parameter set were changed, and a misspelt key raises instead of being ignored.

All returned objects are plain attribute containers with an `as_dict` method that downsamples onto a common index set, which is what the web service serialises. Operating points are cached by `(bias, T, N, approximation)` so that repeated queries from an interactive session cost nothing.

2.3. Browser front end

The same solver is served by a FastAPI application (`obpds-web`, or `uvicorn web.app:app`). The service exposes the presets, accepts a device description as JSON, runs the solve in a worker thread and returns the payload the front end draws: band edges per valley, quasi-Fermi levels,

carrier densities, field, the split valence edges when the stack is strained, subband levels and wavefunctions when the quantum correction is on, and the terminal current with its spread. The front end is deliberately dependency-free—SVG drawn by hand, no framework—because the point of the workbench is that a student can open a browser and see a band diagram without installing anything. The service is not required for any of the numerics; it imports the library and calls the same public methods as the script above.

A concrete exercise, of the kind the workbench was built for: load the AlGaAs/GaAs HEMT preset with the quantum box ticked, then widen the well from 8 nm to 12 nm and watch E_0 fall and the centroid move into the well; or leave the geometry alone and raise the subband count, which shows how many states a well of that depth actually binds before the levels cross E_{cut}. Neither needs an installation, and neither is easy to convey in a static figure.

3. Physical models

3.1. Electrostatics and carrier statistics

With the electrostatic potential ψ in volts (numerically equal to the potential energy of an electron in eV), the Poisson equation is

$$\frac{d}{dx}\left(\varepsilon(x)\frac{d\psi}{dx}\right) = -q\left[p - n + N_{\mathrm{net}}(x)\right], \tag{1}$$

with $N_{\mathrm{net}} = N_D^{+} - N_A^{-}$ and Dirichlet conditions at the contacts. The electron density is summed over the Γ, L and X valleys. For the Γ valley the Kane dispersion is used [10], so that

$$n_\Gamma = \frac{2}{\sqrt{\pi}} N_c^\Gamma \, H_{1/2}(\phi, \alpha), \qquad H_{1/2} = G_{1/2} + 2\alpha\, G_1, \tag{2}$$

where $G_k(\phi,\alpha) = \int_0^\infty t^k \sqrt{1+\alpha t}\,[1+e^{t-\phi}]^{-1} dt$ and $\alpha = (kT/E_g)(1-m^*/m_0)^2$ is the non-parabolicity parameter on the kT scale. The L and X valleys are treated as parabolic with their own densities of states and their own band edges, so that a material that becomes indirect under strain or alloying moves its electrons between valleys without a discontinuity. The holes use the heavy-hole, light-hole and split-off masses, with the split-off band separated

by Δ_{SO}; under strain the three edges are no longer degenerate and the density of states is reweighted (Section 3.2).

The integrals in Eq. (2) are evaluated by Gauss–Legendre quadrature after the substitution $t = u^2$, which removes the $\sqrt{t}$ endpoint singularity in the derivative of the integrand; with 128 nodes and an upper limit $u_{\max} = \sqrt{\max(\phi, 0) + 60}$ the values and their derivatives with respect to ϕ come out of one pass over the nodes, which is what the Newton Jacobian needs. The piecewise minimax rational approximations of Fukushima [11] are more accurate still, but they carry a large coefficient table for a gain that is invisible at the tolerance of a Newton iteration. Against adaptive quadrature the median relative difference of the quadrature is 1×10^{-13} over $-5 \le \phi \le 15$; the single outlier, 8×10^{-10} at $\phi = -5$, is limited by the absolute tolerance of the reference integral, whose value there is only 5×10^{-3}.

The band parameters are those of the Vurgaftman compilation [12], the dielectric constants those of Adachi [13], and the temperature dependence of the gap is Varshni's. A ternary is interpolated between its binaries with a bowing parameter applied to the gap at the device temperature rather than at zero temperature—a detail that moves $\mathrm{In_{0.53}Ga_{0.47}As}$ by 0.1 eV if it is got wrong.

3.2. Pseudomorphic strain

A layer thinner than its critical thickness cannot relax: it takes the in-plane lattice constant of the substrate and compensates along the growth direction. For growth along [001] and a free surface ($\sigma_{zz} = 0$),

$$\varepsilon_{\parallel} = \frac{a_{\mathrm{sub}} - a}{a}, \qquad \varepsilon_{\perp} = -2\frac{C_{12}}{C_{11}}\varepsilon_{\parallel}, \tag{3}$$

so that $\mathrm{Tr}(\varepsilon) = 2\varepsilon_{\parallel} + \varepsilon_{\perp}$. The conduction band moves by the hydrostatic term, $\delta E_c = a_c \,\mathrm{Tr}(\varepsilon)$, for all three valleys.

In the unstrained crystal the heavy- and light-hole edges coincide, so any strain with a shear component resolves a degeneracy instead of perturbing a single level. For a [001] biaxial strain the Bir–Pikus Hamiltonian splits into a diagonal $|3/2, \pm 3/2\rangle$ block and a 2×2 block that mixes $|3/2, \pm 1/2\rangle$ with $|1/2, \pm 1/2\rangle$ through the spin–orbit coupling, and both diagonalise by hand. With

$$h = a_v \mathrm{Tr}(\varepsilon), \quad d = -2b(\varepsilon_{\perp} - \varepsilon_{\parallel}), \quad s = \sqrt{\Delta^2 - \Delta d + \tfrac{9}{4}d^2}, \tag{4}$$

they are

$$E_{\mathrm{hh}} = E_v + h + \frac{d}{2}, \quad E_{\mathrm{lh}} = E_v + h - \frac{\Delta}{2} - \frac{d}{4} + \frac{s}{2}, \quad E_{\mathrm{so}} = E_v + h - \frac{\Delta}{2} - \frac{d}{4} - \frac{s}{2}. \tag{5}$$

The closed form is worth having because it is checkable: at zero strain it reduces exactly to $E_{\mathrm{hh}} = E_{\mathrm{lh}} = E_v$ and $E_{\mathrm{so}} = E_v - \Delta$; the shear part is traceless, so the sum of the three edges is independent of d; and as $\Delta \to \infty$ the heavy–light separation tends to d with an error that falls as $1/\Delta$. All three are assertions in the test suite (Section 5).

Sign conventions for the deformation potentials are a standing trap. We use the Van de Walle–Chuang form [14, 7] ($a_v > 0$, gap shift $(a_c - a_v)\mathrm{Tr}\,\varepsilon$); the Vurgaftman compilation [12] tabulates a_v with the opposite sign and therefore writes the same shift as $(a_c + a_v)$. The parameter file declares which convention it follows, and a regression test pins the hydrostatic coefficient $dE_g/dP = -(a_c - a_v)/B$ at 11.04 meV/kbar for GaAs, within 0.3 meV/kbar of the measured value.

Strain is applied only where it is defined: a layer with no deformation potential in the parameter set (the nitrides) is still reported as strained elastically, but its band edges do not move, and the layer table says so instead of silently doing nothing. Layers thicker than the Matthews–Blakeslee critical thickness [15] are flagged, not relaxed.

3.3. Drift–diffusion transport

The current is written in the thermodynamic form $J_n = q\mu_n n\, dF_n/dx$ and $J_p = q\mu_p p\, dF_p/dx$, with F_n and F_p the quasi-Fermi levels in eV [16, 17]. Continuity is discretised in finite-volume form over a cell of width h,

$$J_n|_{i+1/2} - J_n|_{i-1/2} = +qR_i h, \qquad J_p|_{i+1/2} - J_p|_{i-1/2} = -qR_i h, \tag{6}$$

with the link fluxes given in Section 4.2. Recombination is Shockley–Read–Hall through mid-gap traps plus band-to-band radiative; Auger is not included. Both quasi-Fermi levels are pinned to the metal level at each contact, which is the ideal ohmic contact with infinite surface recombination.

Two details of the recombination term decide whether it is right. The driving force is $np - n_0 p_0$, the departure from equilibrium, and not $np - n_i^2$: under Fermi–Dirac or Kane statistics np differs from n_i^2 by a few per cent even at equilibrium, and that residual appears as a spurious generation rate at zero bias. The trap densities are $n_1 = n_{\mathrm{eff}} e^{(E_t - E_i)/kT}$ and $p_1 =$

$n_{\mathrm{eff}} e^{(E_i - E_t)/kT}$ with $n_{\mathrm{eff}} = \sqrt{n_0 p_0}$, which generalises n_i to whatever statistics are in force; they are *not* the local equilibrium densities. Substituting n_0 and p_0 there is the tempting shortcut—it also makes R vanish at equilibrium—but it is equivalent to letting the trap energy slide with the Fermi level, and in the depletion region it inflates the denominator by the majority density, suppressing generation by two orders of magnitude. The mistake hides well, because in the neutral regions both choices reduce to the same minority lifetime and the forward knee looks correct; only the reverse characteristic exposes it.

Mobilities are the tabulated room-temperature low-field values, scaled as $(T/300)^{-1.5}$. There is no velocity saturation and no doping dependence: those would need parameters that no III–V compilation provides consistently, and inventing them would make the current look more authoritative than it is.

3.4. Self-consistent quantisation

Classical electrostatics puts the electron gas exactly at the band-edge minimum, which for a heterojunction is the interface plane. Confining a state to a slab of thickness L costs at least $\hbar^2\pi^2/2m^*L^2$ of kinetic energy, so the ground state floats above the band edge and its wavefunction has a lobe pushed away from the interface. A position-dependent mass makes the operator itself a choice, and only one arrangement of it is Hermitian: the BenDaniel–Duke form [18]

$$-\frac{\hbar^2}{2}\frac{d}{dx}\left(\frac{1}{m^*(x)}\frac{d\varphi}{dx}\right) + E_c(x)\varphi = E\varphi, \tag{7}$$

It conserves the flux $(1/m^*)d\varphi/dx$ through an interface, which the naive $-\hbar^2\varphi''/2m^* + E_c\varphi$ does not. Discretised on the Poisson grid, with $1/m^*$ averaged harmonically onto the faces, Eq. (7) is a symmetric tridiagonal eigenproblem.

A bound subband is a two-dimensional gas, so its sheet density is closed form, because the two-dimensional Fermi–Dirac integral is a logarithm:

$$N_i = \frac{m^* kT}{\pi\hbar^2}\ln\!\left[1 + e^{(F - E_i)/kT}\right], \tag{8}$$

which stays exact in the degenerate limit. The volume density follows as $n(x) = \sum_i N_i |\varphi_i(x)|^2$. States above the continuum threshold E_{cut}—the lower of the two contact band edges, $E_c(0)$ and $E_c(L)$, since an electron above it

can escape to a contact—are not discarded: their contribution is the three-dimensional density weighted by the regularised incomplete gamma function,

$$n_{\mathrm{cont}}(x) = n_{\mathrm{3D}}(x)\, Q\!\left(\tfrac{3}{2}, \frac{\max(0, E_{\mathrm{cut}} - E_c(x))}{kT}\right), \tag{9}$$

which removes from the classical density exactly the part that the bound states already account for and tends to unity far from the well. Without this term the density outside the well would vanish.

4. Discretisation and solvers

4.1. Finite-volume Poisson

Equation (1) is discretised in flux form on a node-centred uniform grid, with the face permittivity taken as the harmonic mean of the two adjacent nodes,

$$\varepsilon_{i+1/2} = \frac{2\varepsilon_i \varepsilon_{i+1}}{\varepsilon_i + \varepsilon_{i+1}}. \tag{10}$$

This is the exact face value for a piecewise-constant ε whose jump lies between the two nodes, and it is what makes the discrete operator faithful to the unexpanded form of Eq. (1). Writing the equation as $\varepsilon\psi'' + \varepsilon'\psi' = -q\rho$ and then dropping the second term, which is a common simplification, leaves a spurious field at every heterointerface with a permittivity contrast; the harmonic mean retains it without any special-casing.

The nonlinear system is solved by Newton's method with the damping of Bank and Rose [19]: the undamped step is computed, the residual ratio $\xi = \|G(u + s\,\delta u)\|/\|G(u)\|$ is formed, the step is halved if the reduction is smaller than τs, and otherwise accepted with the damping factor updated as $s/(s + (1-s)\xi/2)$. The initial guess is not $\psi = 0$: each node is first solved for local charge neutrality with a scalar Newton iteration, seeded from the Boltzmann expression and refined against the full statistics, so that degeneracy, satellite valleys and non-parabolicity are already accounted for before the global solve starts.

4.2. A Scharfetter–Gummel flux that is exact for any statistics

Exponential fitting requires the density to be exponential in a potential, and the textbook derivation [20] assumes Boltzmann statistics. Rather than accept that restriction, define at every node the generalised potential

$$\Psi_n \equiv F_n - kT \ln \frac{n}{N_{\mathrm{ref}}}, \qquad \Psi_p \equiv F_p + kT \ln \frac{p}{N_{\mathrm{ref}}}, \tag{11}$$

which is an identity for any statistics—parabolic Fermi–Dirac, Kane, multi-valley—because it is built from the solved n and p themselves. In the non-degenerate limit it reduces to $\Psi_n \to E_c - kT\ln(N_c/N_{\mathrm{ref}})$. Because it is defined this way it carries the electrostatic potential, every heterojunction band-edge step, the density-of-states gradients and the degeneracy correction at once, and there is no separate band-gap-narrowing or effective-field term to get wrong.

With $u = (\Psi_{i+1} - \Psi_i)/kT$ the standard link flux is

$$J_n|_{i+1/2} = \frac{q\mu_n kT}{h}\left[n_{i+1}B(-u_n) - n_i B(u_n)\right], \qquad B(x) = \frac{x}{e^x - 1}. \tag{12}$$

Written this way the two terms are each of order $\mu kTn/h$ and cancel to order $\mu n\, dF/dx$; at equilibrium they differ in the sixteenth digit, which for $\mu = 8500\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$, $n = 10^{17}$ cm^{-3} and $h = 10\,\mathrm{nm}$ is a spurious current of 10^{10} A/cm^2. Substituting Eq. (11) into Eq. (12) and using $u_n = w_n - \ln(n_{i+1}/n_i)$ with $w_n = (F_{n,i+1} - F_{n,i})/kT$, the bracket collapses analytically:

$$J_n|_{i+1/2} = \frac{q\mu_n kT}{h}\, n_{i+1}B(-u_n)\left(1 - e^{-w_n}\right), \tag{13}$$

$$J_p|_{i+1/2} = \frac{q\mu_p kT}{h}\, p_i B(-u_p)\left(1 - e^{-w_p}\right). \tag{14}$$

The factor that vanishes at equilibrium is now computed from the quasi-Fermi level itself, so $J = 0$ exactly at $w = 0$, in floating point, with no cancellation involved. This is the form the program uses. The identity between Eq. (12) and Eqs. (13)–(14) is algebraic and holds for any n_{i+1}/n_i, which is what makes it safe for degenerate and non-parabolic carriers.

4.3. Continuity, damping and row equilibration

Each continuity equation is solved by Newton's method for one quasi-Fermi level with the other frozen. The Jacobian is tridiagonal and is assembled analytically, differentiating Eq. (13) with $B(-u)$ lagged—exact for Boltzmann statistics and a good approximation otherwise, with the converged solution checked by residual rather than by the linearisation.

Semiconductor Jacobians span absurd dynamic ranges: in the continuity equations a neutral-region row is of order $\mu kTn/h \sim 10^{27}$ while a Dirichlet row is of order one, so the matrix is numerically singular although it is well conditioned after equilibration. Since Newton's step is invariant under left

multiplication, dividing row i of both the Jacobian and the residual by the row infinity norm is a free preconditioner, and the program does it. This matters more than it sounds: an earlier version measured the residual after equilibration and declared convergence while the minority carrier profile was flat and the current missing, because a residual that looks tight in equilibrated units can still permit an imbalance a billion times the quantity being solved for. The residual tolerance is therefore expressed in physical units, at 10^{-16} of the equilibrated scale, which keeps the current imbalance four decades below the current itself.

4.4. The Gummel loops

Poisson, the two continuity equations and (optionally) the eigenvalue problem are coupled by Gummel iteration [21]. For transport, one sweep is: solve Poisson with the quasi-Fermi levels frozen, then electron continuity, then hole continuity; repeat until the largest change in ψ, F_n or F_p falls below 10^{-7} eV.

The starting point is the part that decides whether this converges at all. In a quasi-neutral region the minority density obeys $Dc'' = (c - c_0)/\tau$, and since the region is thin compared with the diffusion length the profile is very nearly linear between the injected value $c_0 e^{V/kT}$ at the depletion edge and c_0 at the contact. Writing $c \propto e^{\pm F/kT}$ turns that straight line into a logarithmic quasi-Fermi level,

$$F(s) = F_{\text{majority}} + kT \ln\!\left[e^{V/kT}(1 - s) + s\right], \tag{15}$$

with s the fractional distance from the junction to the contact. The expression is exact for either sign of V, because $e^{V/kT}$ is positive however negative V becomes; a formula built on $1 - e^{-|V|/kT}$ instead diverges as soon as the bias changes sign. Getting this shape right matters far more than it looks: a linear ramp—the obvious guess—sits about 0.13 V away from the answer in the middle of the neutral region, and the continuity Jacobian there is a near-Laplacian whose row sum is only $\sim 10^{-4}$, so that residual is amplified into a 100 V Newton step along a near-null direction and the solve is destroyed.

There is deliberately no bias continuation. Walking up from equilibrium looks like the safe thing to do, but an ohmic contact pins the carrier *densities*, not the levels: reusing the previous operating point at a new bias leaves the contact densities off by $e^{\Delta V/kT}$ and the first nonlinear Poisson solve diverges. With the analytic start of Eq. (15) a bias point converges in two or three

sweeps from cold, so there is nothing for continuation to save, and every bias point is independent—which also makes an I–V sweep trivially parallel.

The quantum loop is the same idea with the density replaced: solve the eigenproblem on the current band edge, rebuild n from Eqs. (8) and (9), re-solve Poisson with that density held fixed, repeat. Holding the density fixed is what makes this a Gummel iteration rather than a coupled Newton solve; the price is linear convergence, the reward is that each step is a well-conditioned Poisson problem. Undamped, the map turns out to be a contraction, with the potential change falling by a factor of about 0.78 per sweep, because the feedback is self-limiting: pushing electrons off the interface lowers the field there, which shallows the well, which pushes them back. Damping is kept only as a safety net, applied when a sweep makes the residual worse.

The quantum loop is run at equilibrium only. There is no quantum-corrected drift–diffusion: the I–V curve, the C–V curve and the current of Section 6 are all classical, and the subband structure is a property of the zero-bias band profile.

4.5. The eigenproblem

Three details of the discretisation decide whether the spectrum is right, and none of them shows up in a plot of the wavefunction.

Walls. The zero condition is imposed on a ghost node one step outside the grid. Its value never enters the matrix, but its flux does, on the diagonal of the first and last rows. Dropping that term makes the wall reflecting: every level falls by about one mesh step and a state appears near $E \approx 0$, so a triangular well returns less than half its true ground-state energy.

Inner product. The assembled operator is self-adjoint under the nodal sum, which fixes the normalisation to $\|\varphi\| = (\sum_i \varphi_i^2)^{1/2}\sqrt{h}$. Weighting the nodes trapezoidally belongs to a different inner product and leaves the Gram matrix orthogonal only to 10^{-6}, against 10^{-16} for the nodal sum.

Interface mass. The mass sits inside the derivative in Eq. (7), so two half-cells in series combine harmonically, as the permittivity does in Eq. (10). An arithmetic average of $1/m^*$ is the common shortcut, and it costs the matrix its symmetry.

The eigenvalues come from `eigh_tridiagonal`, a symmetric-tridiagonal solver asked for the lowest few states by index: one tridiagonal factorisation replaces a full spectrum. The density-of-states mass of a subband is the in-plane mass, and a state that straddles an interface samples both materials,

so it is taken as the $|\varphi|^2$-weighted average rather than the mass at whichever node happens to be the well bottom.

4.6. Reading the terminal current

$J_n + J_p$ is constant in exact arithmetic, but its discrete value is not equally trustworthy everywhere. A majority carrier in a quasi-neutral region carries the current as $q\mu c\, dF/dx$ with $c \sim 10^{17}$ cm^{-3}, so the gradient belonging to a 10^{-15} A/cm^2 current is $\sim 10^{-22}$ eV per cell—five orders of magnitude below the round-off of a stored level of order 0.1 eV. Those links report pure noise. Inside the depletion region both carriers are minorities, both densities are near n_i and both fluxes are clean, so the program reads the terminal current as the median of $J_n + J_p$ over the links where $\max(n, p) \le 10^3 n_i$. If no link satisfies that criterion—at high forward bias the depletion region fills with carriers and none does—the median is taken over the whole device, and `J_spread` then states how little the number deserves to be trusted. It reports

$$\texttt{J_spread} = \frac{\max_k |J_k - J|}{|J|}, \tag{16}$$

the relative deviation of the worst link. This is not a cosmetic diagnostic. At 0.8 V forward bias it is 1.9×10^{-4}, at -1 V it is 373: the reverse current of this device is not resolvable beyond its leading digits at $N = 1000$, and the program says so instead of printing six figures.

5. Verification

5.1. Strategy

The program is held to closed-form results, not to itself. Three suites and an end-to-end API test carry 196 assertions (Table 2); Figure 2 plots the achieved relative error of each check against the tolerance it is held to. Every tolerance is stated in the test source, and every reference value is either an analytic expression or a literature number with its provenance in a comment.

5.2. Classical electrostatics and transport

For a GaAs p–n junction with 10^{17} cm^{-3} on both sides ($n_i = 1.70 \times 10^6$ cm^{-3} at 300 K) the built-in potential is reproduced to 8.7×10^{-4} relative, the depletion width inferred from $2V_{\mathrm{bi}}/E_{\max}$ to 2.7%, and the peak field to 2.6%; the deviations are those of the depletion approximation itself, which

Table 2: Verification suites. Wall times are for one core of the machine described in Section 7.

Suite	Checks	Time	Scope
`validate_physics.py`	13	206 s	depletion approximation, short-base diode law, ideality factor, $1/\tau$ scaling, zero current
`validate_quantum.py`	31	21 s	square, triangular and mass-step wells, 2D population limits, self-consistent loop
`validate_strain.py`	71	0.9 s	strain tensor, Bir–Pikus identities, pressure coefficient, critical thickness
`web_smoke.py`	81	37 s	API contract, every preset, every payload field
Total	196	265 s	

the simulation is not obliged to obey. The integral of the net charge over the device is 1.5×10^{-21} C/cm^2.

For the current, the checks are the ones a diode obeys. The short-base diode law $J = qn_i^2(D_n/W_pN_A + D_p/W_nN_D)e^{V/kT}$ is reproduced to 6.3%, 4.7% and 4.0% at 0.80 V, 0.85 V and 0.90 V; the local ideality factor is 1.01 in the diffusion regime and 1.92 in the SRH-dominated regime, the textbook two-region crossover; the reverse current scales as $1/\tau$ with the ratio 9.99997 for a decade change in lifetime; and the equilibrium current is exactly zero. Current conservation at 0.8 V holds to 1.9×10^{-4}, which is the `J_spread` of that operating point. The reverse generation current at -0.5 V is 5.3×10^{-12} A/cm^2 against 8.7×10^{-12} A/cm^2 from the generation-volume estimate, a 39% agreement—as good as that estimate, which assumes a sharp depletion edge, can be.

The Kane statistics are checked where they matter: for InAs ($\alpha = 0.070$) the non-parabolic integral exceeds the parabolic one by 28.7% at $\phi = 2$, and the three approximations (Boltzmann, parabolic Fermi–Dirac, Kane) agree in the non-degenerate limit and diverge as they should as the density approaches the band.

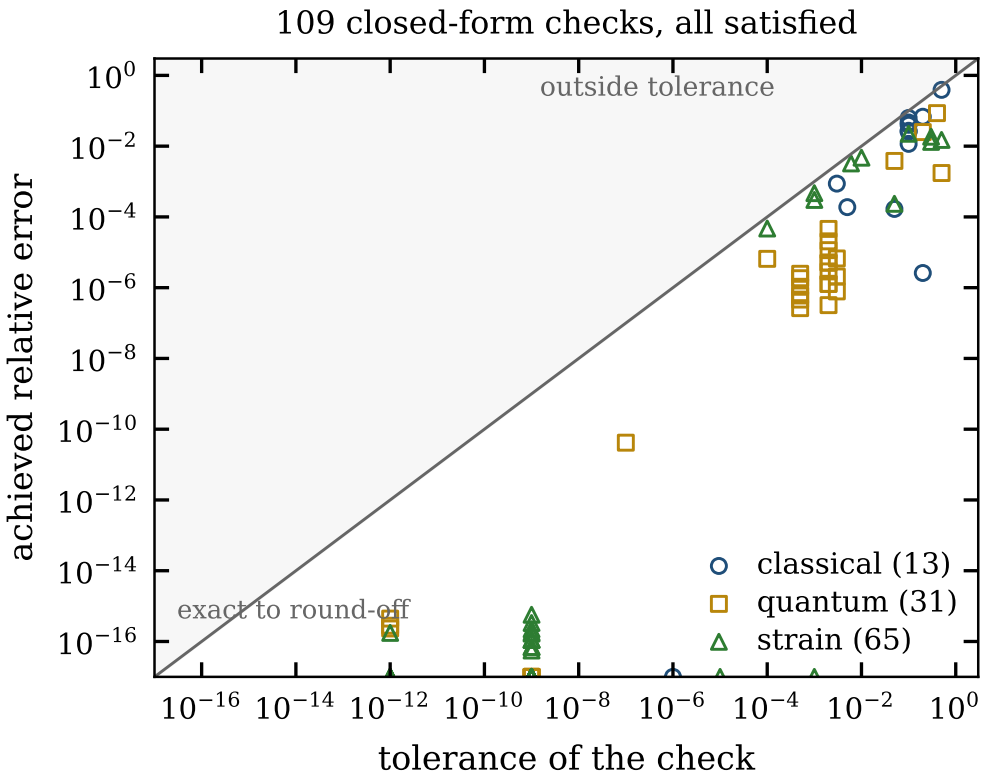


Figure 2: Achieved relative error of each closed-form check against the tolerance it is held to, for the three physics suites. Every point lies below the diagonal; the cluster on the bottom axis is exact to round-off. The 81 API assertions of `web_smoke.py` check payload consistency rather than physics and are not shown.

5.3. Quantisation

An infinite square well reproduces $\hbar^2\pi^2n^2/2m^*L^2$ to 3×10^{-7} at $N = 1601$, with second-order convergence in the mesh. A triangular well reproduces the Airy zeros to 2.6×10^{-7} once the analytic reference is evaluated at the position of the ghost wall, $x = -h$, the residual difference from evaluating it at $x = 0$ is a constant $Fh = 0.1\,\mathrm{meV}$, a property of the discretisation. The mass-step well is the check that actually tests the variable-mass form: the solver reproduces the transcendental condition obtained from Eq. (7) with the BenDaniel–Duke interface condition to 7×10^{-7}, whereas a one-sided numerical derivative is only first-order accurate and cannot distinguish the

correct formulation from a wrong one. The eigenfunctions are orthonormal to 4×10^{-16}.

The two-dimensional population, Eq. (8), is checked in both limits: 4×10^{-11} against the Boltzmann expression and 7×10^{-6} against the degenerate asymptotic form.

5.4. Strain

The three identities of Section 3.2 are asserted for four materials at four strain values: the zero-strain limit is exact to 10^{-12}, the shear part is traceless to 6×10^{-16} eV, and the $\Delta \to \infty$ limit converges to d with the measured order 2.00 in $1/\Delta$ (errors 4.7×10^{-3}, 4.7×10^{-4}, 4.7×10^{-5} for $\Delta = 10$, 100, 1000 eV). The hydrostatic pressure coefficient lands at 11.04 meV/kbar for GaAs and 10.49 meV/kbar for InAs. $In_{0.53}Ga_{0.47}As$ on InP is lattice matched to 3×10^{-4}, and the critical thickness decreases monotonically with mismatch, from 41.3 nm at 0.5% to 1.98 nm at 4%.

The calibration interface of Section 2 is checked in the same suite, because a half-applied parameter change is worse than none: shifting a layer's valence-band offset by 0.1 eV moves its conduction edges by exactly the same amount and leaves the gap untouched to 10^{-12} eV; overriding a_c to zero removes the hydrostatic shift and nothing else; and a misspelt key, or one that would leave derived quantities stale, raises instead of being silently ignored.

5.5. Negative controls

Three checks are there to catch a correction that adds something where nothing should be added, which is the failure mode of a quantisation scheme.

A classical junction must come back unchanged. On a GaAs p–n junction the quantum and classical densities agree to zero—bit for bit, at $N = 400$—and the solver finds no bound state at all, because E_{cut} lies below every level. If the subband and continuum pieces double counted or dropped density, this test would fail immediately.

A layer on its own substrate must not be strained. Setting the substrate to the layer material gives zero strain, zero valence shift and an unchanged band offset.

The equilibrium current must vanish identically. It does, by construction of Eq. (13), and the assertion is kept because the construction is easy to break.

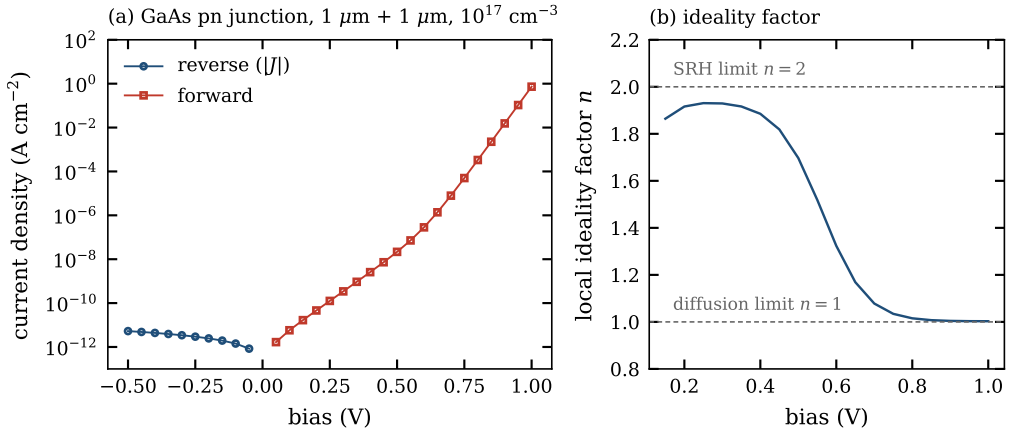


Figure 3: (a) Current density of the GaAs p–n junction against bias. (b) Local ideality factor, showing the SRH-limited and diffusion-limited regimes. The values at the two plateaux are 1.92 and 1.01.

5.6. The API contract

The 81 assertions of `web_smoke.py` post every preset through the service and check the payload the front end actually draws: that each field the browser reads is present and finite, that the arrays share one index set, that the subband populations sum to the reported sheet density, that one wavefunction is returned per bound state and that its length matches the grid, and that the electron column contains the quantum density rather than the classical one. This is the class of bug that a physics test cannot catch and a user notices immediately, and it is cheap to keep covered.

6. Representative calculations

6.1. A p–n diode

Figure 3 shows the I–V characteristic of the GaAs junction used above, over eight decades of current, with the local ideality factor $n = (q/kT)^{-1} dV/d\ln J$. The two regimes are cleanly separated: $n \simeq 1.9$ below 0.35 V, where recombination in the depletion region dominates, and $n \simeq 1.01$ above 0.7 V, where diffusion in the neutral regions does. The reverse current is flat to within the noise floor discussed in Section 7—a feature of the model (no avalanche, no tunnelling) as much as of the solver.

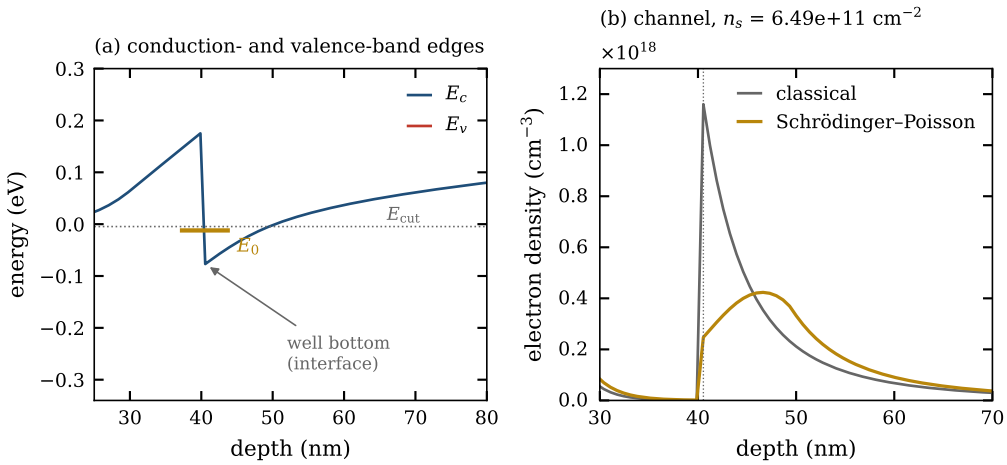


Figure 4: (a) Band edges of the modulation-doped heterojunction with the bound subband marked; E_{cut} is the continuum threshold of Eq. (9). (b) Electron density near the interface: the classical solution peaks at the potential minimum, the Schrödinger–Poisson solution peaks 4.6 nm behind it.

6.2. A modulation-doped heterojunction

The second structure is the one the quantum correction exists for: 30 nm of $Al_{0.3}Ga_{0.7}As$ doped at 10^{18} cm^{-3}, a 10 nm undoped spacer, and 500 nm of GaAs. Donors in the barrier spill electrons into the GaAs, where they collect in a notch narrow enough to quantise.

At $N = 800$ the self-consistent loop converges to a single bound subband with $E_0 - E_{c,\min} = 64.8$ meV, a sheet density of 6.49×10^{11} cm^{-2}, and a $|\varphi_0|^2$ centroid 4.6 nm behind the potential minimum (Figure 4). The classical solution puts its peak at the minimum itself. The subband energy sits 25 % below the bare-triangular estimate at the same field (86.3 meV at $F = 9.4 \times 10^4$ V/cm), which is the expected direction: the well is not a bare triangle, because the charge that fills it flattens the field.

The total electron sheet density of the structure is 2.9×10^{12} cm^{-2}, of which only 6.5×10^{11} cm^{-2} is in the channel; the rest remains in the doped barrier. That is the correct physics for a single heterojunction with a 30 nm doped barrier, and it is a reminder that the sheet density reported for a "2DEG" is only the channel part if the structure is designed so.

6.3. A strained well

An $In_{0.2}Ga_{0.8}As$ layer on GaAs is mismatched by +1.43%; the wafer locks it to $\varepsilon_{\parallel} = -1.413\%$ in plane, and it answers with $\varepsilon_{\perp} = +1.317\%$ along the growth direction. The valence edge splits by 78.9 meV, heavy hole up. The bare shear parameter is $d = 93.9$ meV, so a calculation that left the spin–orbit coupling out would put the light hole 19 % too low. The critical thickness is 10.15 nm: an 8 nm well is pseudomorphic, a 20 nm one is flagged. The gap opens from 1.1324 eV to 1.2043 eV and the conduction edge rises by 101.9 meV, both of which move the confinement.

7. Performance and the limits of the numbers

7.1. Cost

Figure 5 reports wall times on one core of an Intel Core i7-12700, Windows 11, Python 3.13.5, NumPy 2.5.4, SciPy 1.18.1. The nonlinear Poisson solve at $N = 1000$ takes 0.35 s and scales between linear and quadratic in N; a drift–diffusion bias point takes 1.5 s in two Gummel sweeps; a full I–V point, cold-started and including the harder reverse and high-forward biases, averages 6.8 s; and a quantum equilibrium solve for the heterojunction takes 31.8 s on the run recorded in the figure, with repeated runs of the same script spread between 18 s and 32 s. These are single-core numbers on a problem of a few thousand unknowns; nothing here needs a cluster, and an I–V sweep parallelises trivially because the bias points are independent.

Almost all of the quantum cost sits in one kernel. A profile of the $N = 800$ case puts 82% of the runtime in the Fermi–Dirac quadrature: 7956 evaluations of four integrals over 128 nodes, for 58 Poisson solves. The node count is deliberate—64 nodes would halve the time but is only good to 6×10^{-6} relative, and a quadrature error propagates straight into the current. Batching the three valleys into one call and tabulating $H_{1/2}$ in ϕ are the obvious levers and are left for a later version. The timings, the machine and the library versions are written by `benchmark.py` into the JSON that generated Fig. 5, so each number is traceable to a specific run.

7.2. Accuracy limits, and saying so

Two numerical limits are worth stating explicitly, because both are invisible in a plot and both are properties of the discretisation rather than of the physics.

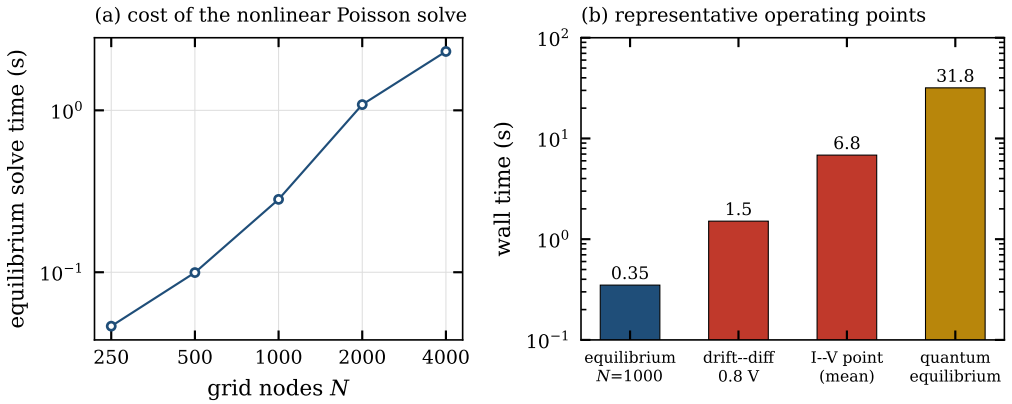


Figure 5: (a) Cost of the nonlinear Poisson solve against the number of grid nodes. (b) Representative operating points. Numbers above the bars are seconds.

The first is the current noise floor. Figure 6 shows the reported current against the worst discrete link at each bias. At 0.8 V the spread is 1.9×10^{-4}; at 0.4 V it is 0.42; at -1 V it is 373. The interpretation is straightforward: above about 0.6 V the reported current is meaningful to the digits shown, at 0.4 V only the leading digit is, and below 0.1 V the number is round-off. The program reports the diagnostic because a minority-carrier device at low bias is exactly where a user is most likely to believe a number that the arithmetic cannot support.

The second is the mesh sensitivity of the quantised sheet density. For the heterojunction of Section 6 the sheet density is 7.28×10^{11} cm^{-2} at $N = 400$, 6.49×10^{11} cm^{-2} at $N = 800$ and 6.46×10^{11} cm^{-2} at $N = 1600$: 12% between the two coarsest meshes and 0.5% between the two finest. The default $N = 1000$ is therefore adequate for the sheet density to a few per mille, but a calculation that quotes three digits at $N = 400$ does not. The ground-state energy behaves the same way: -16.8 meV, -12.1 meV, -11.9 meV at the same three meshes, because a coarser grid widens the effective well by roughly one cell.

7.3. What the model does not contain

The following are outside the model, and are listed so that the numbers above are read at their proper scope. The model is one-dimensional and isothermal; the transport is drift–diffusion with constant low-field mobilities, with no velocity saturation, no inter- or intra-valley scattering and no

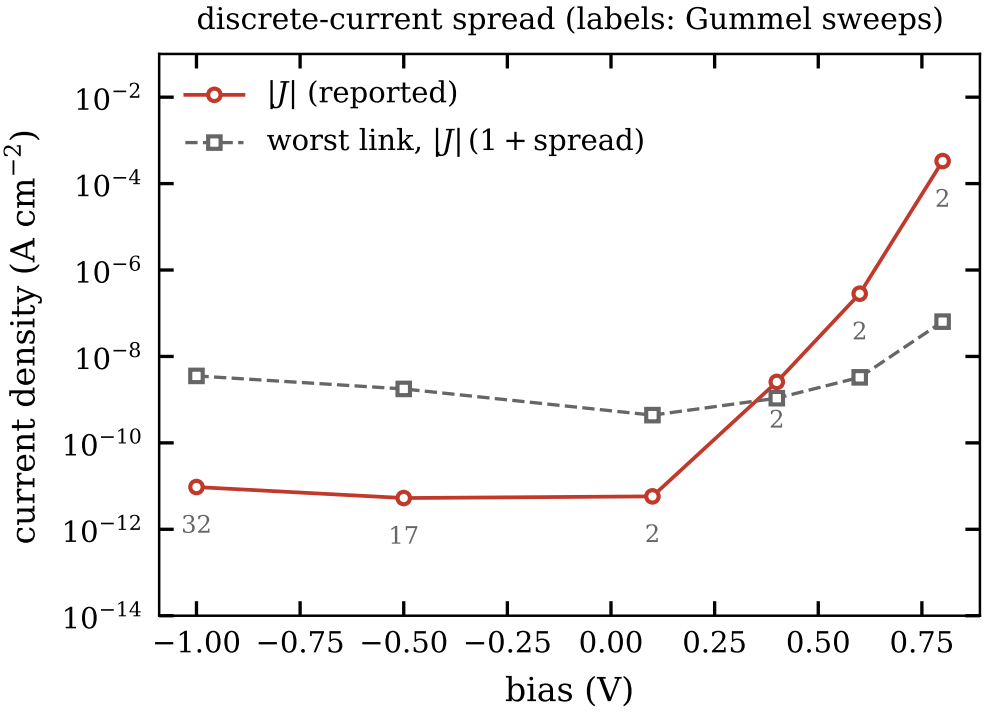


Figure 6: Reported current density and the worst discrete link against bias for the GaAs p–n junction at $N = 1000$. Labels give the number of Gummel sweeps. Where the two curves converge, the discrete current is consistent; where they separate, the reported value is dominated by round-off in the quasi-neutral regions.

energy transport; holes are not quantised, so the valence-band figures are edge splittings and not valence subbands; the quantum correction is applied at equilibrium only, so no current or capacitance reported here includes it; there is no exchange–correlation correction to the confined gas; strain relaxation is reported, not modelled; the piezoelectric charge is absent, which for [001] growth is exact, because a biaxial strain on that orientation has no shear component; the same a_c is used for the Γ, L and X valleys and the shear deformation potential Ξ_u is omitted; effective masses do not change with strain; and there is no Auger recombination.

One disagreement with experiment is worth recording. For strained $\mathrm{In}_x\mathrm{Ga}_{1-x}\mathrm{As}$ wells our conduction-band offset at $x = 0.10$ is 41 meV, against 82 meV from the capacitance–voltage determination of Zubkov *et al.* [22],

which fits $\Delta E_C = 0.85x - 0.3x^2$ eV. The discrepancy is a factor of two and is systematic in x. Part of it is expected—a C–V offset is extracted with an assumed effective mass and an assumed well width, and the two are correlated with the offset—but a factor of two is too large to attribute to that alone, and the most likely cause is the valence-band offset the model starts from, which sets how much of the strain-induced gap change goes to the conduction band. We report the comparison. The override interface of Section 2 exists for this kind of calibration: a reader who prefers a different valence-band offset can supply it per layer and watch what it does to the offset, the subband energies and the sheet density, without touching the parameter tables.

8. Conclusions

OBPDS 2.0 is a one-dimensional solver for III–V heterostructures that computes band diagrams, terminal currents, capacitances and quantised subband structure on one grid, with pseudomorphic strain as a device-level property. It is a derivative of a program that stopped at the zero-current approximation; the additions are drift–diffusion transport, self-consistent quantisation and strain, and the numerical work that makes them correct in a material system whose carrier densities span more than twenty decades.

The parts we would point a reader to are the ones where the obvious implementation is wrong in a way that looks right. A Scharfetter–Gummel flux written in generalised potentials and rearranged to be cancellation-free gives an exactly zero equilibrium current, where the textbook form gives 10^{10} A/cm^2 of numerical noise. Row equilibration of the continuity Jacobian, with the residual tolerance expressed in physical rather than equilibrated units, is the difference between a converged minority profile and a flat one. A ghost-node flux, a nodal-sum normalisation and a harmonic interface mass are three lines of code that decide whether the spectrum is right. And in III–V materials the terminal current of a quasi-neutral region is below the round-off of the stored levels, so the current has to be read where it is clean and its spread reported.

All of this is held to 196 assertions against closed-form results, including the negative controls that make a quantisation scheme believable. The program runs a representative bias point in about a second, is served by a browser front end for classroom use, and is released under the same licence as the work it derives from.

CRediT authorship contribution statement

First Author: Conceptualization, Methodology, Software, Validation, Writing – original draft. Second Author: Formal analysis, Investigation, Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies

Generative AI tools were used to draft and refactor code and to prepare a first draft of the manuscript text. The authors reviewed and edited the content, ran all calculations, and take full responsibility for the published work.

Data availability

The program source, the verification suites, the scripts that regenerate every figure and table of this article, and the raw output of every run are archived in the CPC Program Library and at https://doi.org/10.5281/zenodo.XXXXXXX under AGPL-3.0-or-later.

Acknowledgements

This work reuses and extends the Open Band Parameters Device Simulator [2], distributed under AGPL-3.0; the derived code is released under the same licence. We thank the original author for making it available.

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